



Enterprise Water Systems Lakehouse

Complete Data Pipeline & Governance Dashboard



Pipeline Status



ETL Pipeline Complete



Bronze Layer Created



Silver Layer Processed



Gold Layer Generated



Visualisations created



8 reports generated



Data Lineage Flow



Data Assets Created

 fct_meter_readings Type: Fact Table Layer: Silver Purpose: Core meter reading facts with asset relationships	 dim_asset Type: Dimension Table Layer: Silver Purpose: Asset master data with hierarchical structure	 nwr_daily_summary Type: Aggregated Fact Layer: Gold Purpose: Daily Non-Revenue Water analytics	 asset_failure_features Type: ML Features Layer: Gold Purpose: Predictive maintenance feature store
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Generated visualisations

Asset Failures By Asset Type

View interactive
image

View

Asset Failure Rate By Age

View interactive
image

View

Monthly Nrwr Trend

View interactive
image

View

Nrwr By Day Of Week

View interactive
image

View

Meter Consumption Distribution

View interactive
image

View

Asset Failure Histogram

View interactive
image

View

Nrwr Timeseries

View interactive
image

View

Hourly Consumption Heatmap

View interactive
image

View

Architecture Components

Spark Cluster

Status: Running

Master: localhost:8080

Purpose: Distributed data processing

MinIO Storage

Status: Running

Console: localhost:9001

Purpose: S3-compatible object storage

Delta Lake

Status: Active

Format: Lakehouse tables with ACID transactions

Purpose: Medallion architecture storage

Enterprise Validation Complete

All Systems Operational

- **Data Pipeline:** Successfully processed Bronze → Silver → Gold layers
- **Storage Layer:** Delta Lake tables created with proper schema evolution
- **Analytics:** Business intelligence reports generated and accessible
- **Governance:** Data lineage tracked through Spark job execution

- **Visualization:** Complete BI suite with interactive dashboards

Result: Enterprise-grade Water Systems Lakehouse is fully operational and ready for production workloads.